

COMPUTER NETWORKS

Course Outline :

Sr. No.	Topic	Actual Teaching Hours
1	Computer Networks and the Internet: Internet basics, protocol, network edge and network core, performance measures: delay, packet loss and throughput	5
2	Application Layer: Principles of network applications, HTTP, HTTP Message format, web caching, File Transfer Protocol, SMTP, DNS basics, Peer-to-Peer file distribution, Socket programming with TCP and UDP	8
3	Transport Layer: Overview of transport layer in the Internet, Transport and Network layer relationship, UDP, UDP checksum, concept of reliable data transfer, Go-Back-N and Selective-Repeat, TCP flow and congestion control	12
4	The Network Layer:	15

P:

	Virtual circuit and datagram network, Internet protocol, IPV4 addressing, IPV6, ICMP, routing, routing algorithms: Link-State and Distance-Vector routing algorithm, Routing in the Internet, RIP, OSPF and BGP	
5	The Link layer: Link layer basics, error detection and correction, parity, checksum and CRC, random access control protocols, taking turns protocols, DOCSIS, Link virtualization, Multiprotocol Label Switching	5
Total		45

Recommended Books

Book	Author	Publisher
An Engineering Approach to Computer Networking	Keshav S.	Pearson Education, ISBN 981-235-986-9
Computer Networks, A 4th Edition	Tanenbaum	, PHI ISBP 81 - 203 - 2175 -8
Data Communications and Networking, 3rd edition	Fourauzan B.	Tata McGraw Hill Publications, 2004, ISBN 0 - 07 -058408 - 7.
Network Security Essentials, 3rd edition	William Stallings	PEARSON

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Sr. No.	Component	CO	Max marks	Weight	Tentative date
1.	Poster Presentation	CO1	5(CO1)	16.66% (CO1)	06/08/2026
2.	Unit test	<u>CO2</u> , <u>CO3</u>	15 (CO2 – 6 Marks, CO3 – 9 Marks)	50% (CO2 - 20%, CO3 - 30%)	07/09/2026 to 12/09/2026
3.	Digital Content Creation (e.g. You Tube video)	CO4, CO5	10 (CO4 – 5 Marks, CO5 – 5 Marks)	33.33 (CO2 – 16.65%, CO3 – 16.65%)	16/10/2026

□

Course Outcomes(CO)

C01	Explain basics of networking, Internet, Networking protocols and layered architecture of networks.
C02	Understand and explore major application layer protocols—DNS, HTTP, FTP, and SMTP—and analyze their operation, architecture, and real-world applications
C03	Apply knowledge of TCP and UDP transport layer protocols to analyze the functioning of TCP flow control, congestion control, and congestion avoidance techniques.
C04	Understand the fundamental concepts of the network layer and Internet Protocol (IP),and describe routing protocols and algorithms used in wired and wireless networks.
C05	Understand and explain key concepts and technologies of the link layer, including error control, access protocols, virtualization, and switching mechanisms.

Industry Applications of Computer Networks

Healthcare

Telemedicine & remote patient monitoring
Hospital information systems
Medical IoT devices

Banking & Finance

Secure online banking
ATM & payment networks
Fraud detection data exchange

Manufacturing

Industrial IoT
Smart factories
Predictive maintenance

Education

Online learning
Virtual labs
Cloud-based collaboration

APPLICATIONS OF COMPUTER NETWORKS IN HEALTHCARE

Electronic Health Records (EHR)



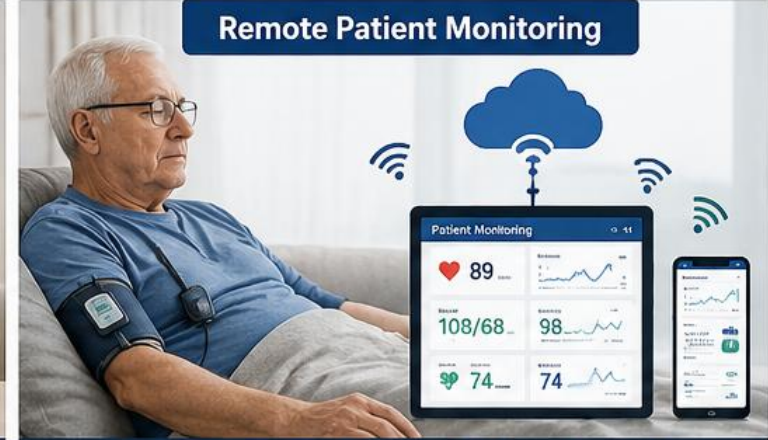
Secure sharing of patient information across hospitals and departments.

Telemedicine & Virtual Consultation



Doctors and patients connect in real time through video, audio and data.

Remote Patient Monitoring



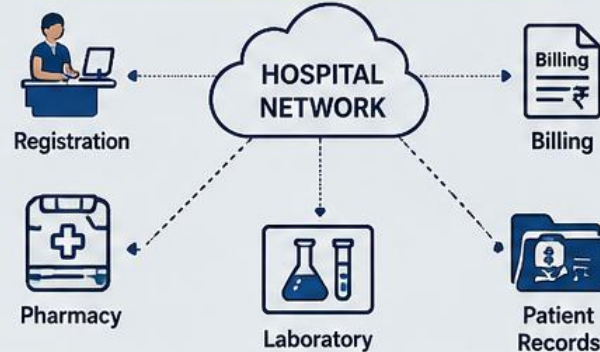
IoT devices collect patient data and send it to healthcare providers over networks.

Medical Imaging & Data Sharing



Images and reports are transferred quickly to specialists for diagnosis.

Hospital Information Systems (HIS)



Different departments connected on a network for efficient management and communication.

Smart Ambulance & Emergency Services



Ambulance sends real-time patient data to hospitals for better and faster treatment.

KEY BENEFITS



Better Patient Care



Faster Diagnosis and Treatment



Secure Data Exchange



Cost Effective



Improved Coordination Among Healthcare Providers

APPLICATIONS OF COMPUTER NETWORKS IN BANKING & FINANCE

1. ONLINE BANKING



Customers can access their accounts, transfer money, pay bills and check balances securely over the Internet.

2. ATM NETWORKS



ATMs are connected to bank servers through secure networks to authorize transactions in real time.

3. MOBILE BANKING & UPI PAYMENTS



Secure real-time money transfers, bill payments and recharges using mobile networks and Internet.

4. ELECTRONIC FUNDS TRANSFER (EFT)



Funds are transferred electronically between accounts across branches and even across countries.

5. STOCK TRADING & INVESTMENT



Stock exchanges and trading platforms use high-speed networks for real-time trading and market updates.

6. CREDIT CARD TRANSACTIONS



Card payments are authorized through secure networks connecting merchants, banks and payment gateways.

7. FRAUD DETECTION & SECURITY



Networks help monitor transactions in real time to detect fraudulent activities and prevent cyber attacks.

8. CORE BANKING SOLUTIONS (CBS)



All branches, ATMs and digital channels are connected to a central system for real-time data synchronization.

9. DIGITAL FINANCE & FINTECH



Fintech services like digital wallets, blockchain, insurtech and lending platforms rely on computer networks.

KEY BENEFITS



Real-time Transactions



Secure Data Transmission



Better Customer Experience



Cost Efficiency



Business Growth & Global Connectivity

APPLICATIONS OF COMPUTER NETWORKS IN MANUFACTURING

1. INDUSTRIAL IoT (IIoT)



Machines, sensors and devices are connected over networks to collect and share data in real time for better monitoring, analysis and decision making.

2. REAL-TIME MONITORING & CONTROL



Networks enable real-time communication between PLCs, SCADA systems and control centers to monitor and control machines instantly.

3. INDUSTRIAL AUTOMATION



Robots, controllers and automated systems communicate over industrial networks (Ethernet/IP, PROFINET, Modbus TCP, etc.) to perform coordinated operations.

4. PREDICTIVE MAINTENANCE



Sensor data is sent over networks to analytics platforms to predict equipment failures and schedule maintenance before breakdowns occur.

5. SUPPLY CHAIN & INVENTORY MANAGEMENT



Networks connect suppliers, manufacturers, warehouses and retailers for real-time tracking of materials, products and shipments across the supply chain.

6. DIGITAL TWIN & SIMULATION



Real-time data from machines is transmitted over networks to create digital twins for simulation, optimization and performance improvement.

7. QUALITY CONTROL



Vision systems and sensors send inspection data over networks for automatic quality checks and immediate alerts on defects.

8. REMOTE ACCESS & SUPPORT



Engineers can securely access machines and systems remotely through networks to troubleshoot, update and provide support.

9. ENERGY MANAGEMENT



Smart meters and devices connected over networks monitor energy consumption and help optimize usage, reducing costs and environmental impact.

KEY BENEFITS



Increased Productivity



Real-Time Visibility



Reduced Downtime



Better Quality & Accuracy



Cost Efficiency



Data-Driven Decisions



Improved Safety & Compliance

APPLICATIONS OF COMPUTER NETWORKS IN EDUCATION

1. E-LEARNING & ONLINE CLASSES



Students and teachers connect through the Internet for live classes, recorded lectures and doubt solving sessions.

2. VIRTUAL LIBRARIES & ONLINE RESOURCES



Networks provide access to digital libraries, e-books, journals and research papers from around the world.

3. LEARNING MANAGEMENT SYSTEMS (LMS)



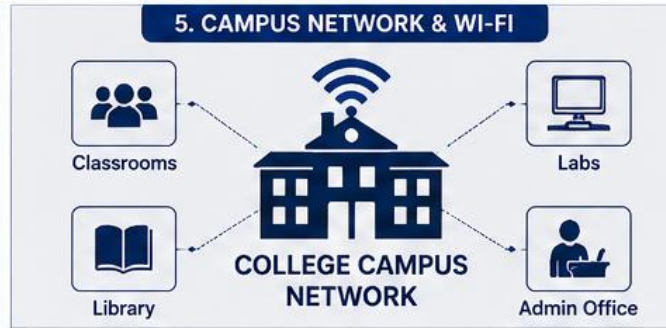
LMS platforms like Moodle, Google Classroom, Canvas use networks to distribute materials, collect assignments and track student progress.

4. ONLINE EXAMINATIONS



Secure networks enable online tests and assessments with real-time monitoring and instant results.

5. CAMPUS NETWORK & WI-FI



Campus networks connect classrooms, labs, libraries and administration for seamless communication and data sharing.

6. CLOUD STORAGE & COLLABORATION



Students and faculty store, share and collaborate on files from anywhere using cloud services (Google Drive, OneDrive, etc.).

7. VIDEO CONFERENCING & WEBINARS



Networks enable real-time interaction through video conferencing, webinars, guest lectures and workshops.

8. SMART CLASSROOMS



Smart classrooms use networked devices, interactive boards and Internet access for engaging and effective learning.

9. COMMUNICATION & NOTIFICATIONS



Networks help send instant notifications about schedules, exams, events and important announcements.

KEY BENEFITS



Anywhere Access



Better Collaboration



Access to Global Resources



Saves Time & Cost



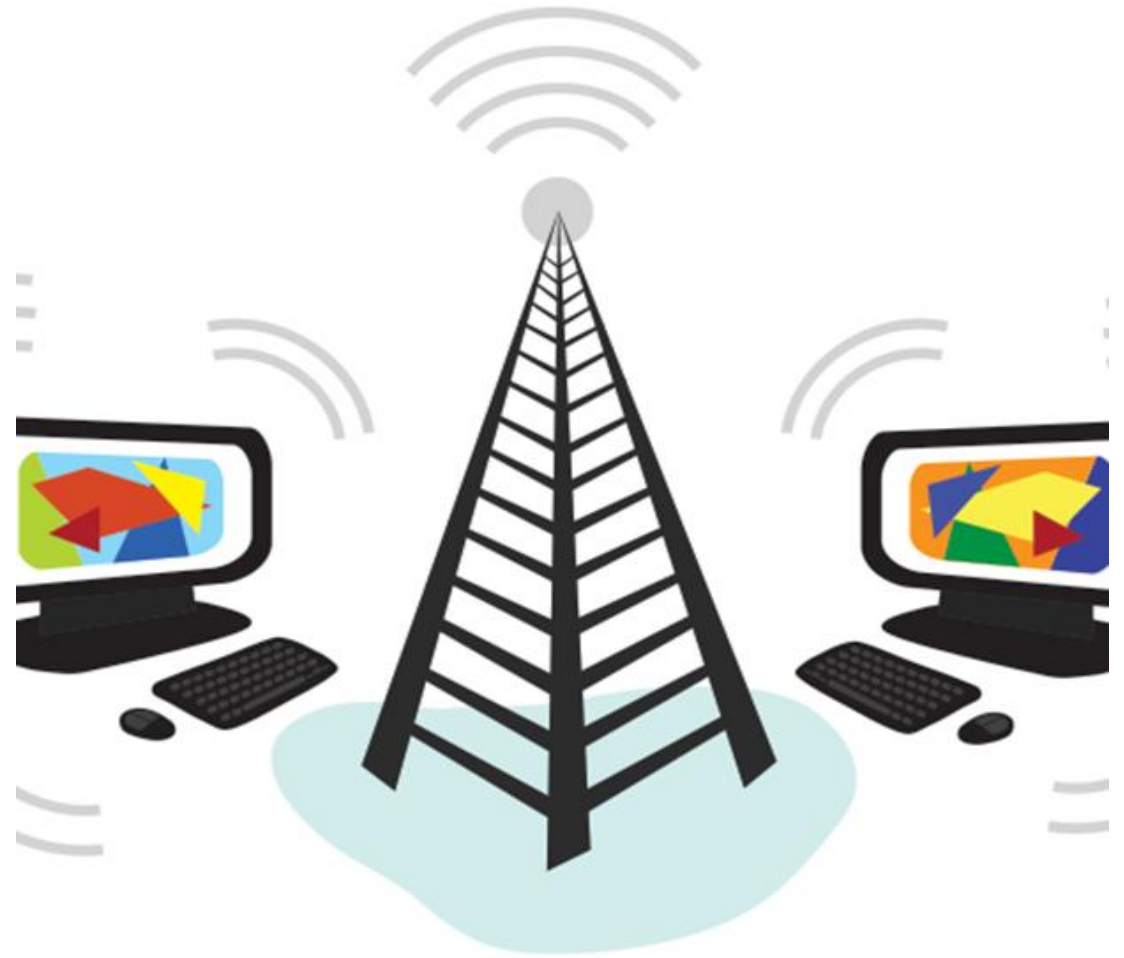
Improved Learning Outcomes



Secure Data & Privacy

What is the Internet?

- The Internet is a **global network** of billions of computers and other electronic devices.
- With the Internet, it's possible to access almost any information, communicate with anyone else in the world, and do much more.
- You can do all of this by connecting a computer to the Internet, which is also called **going online**.

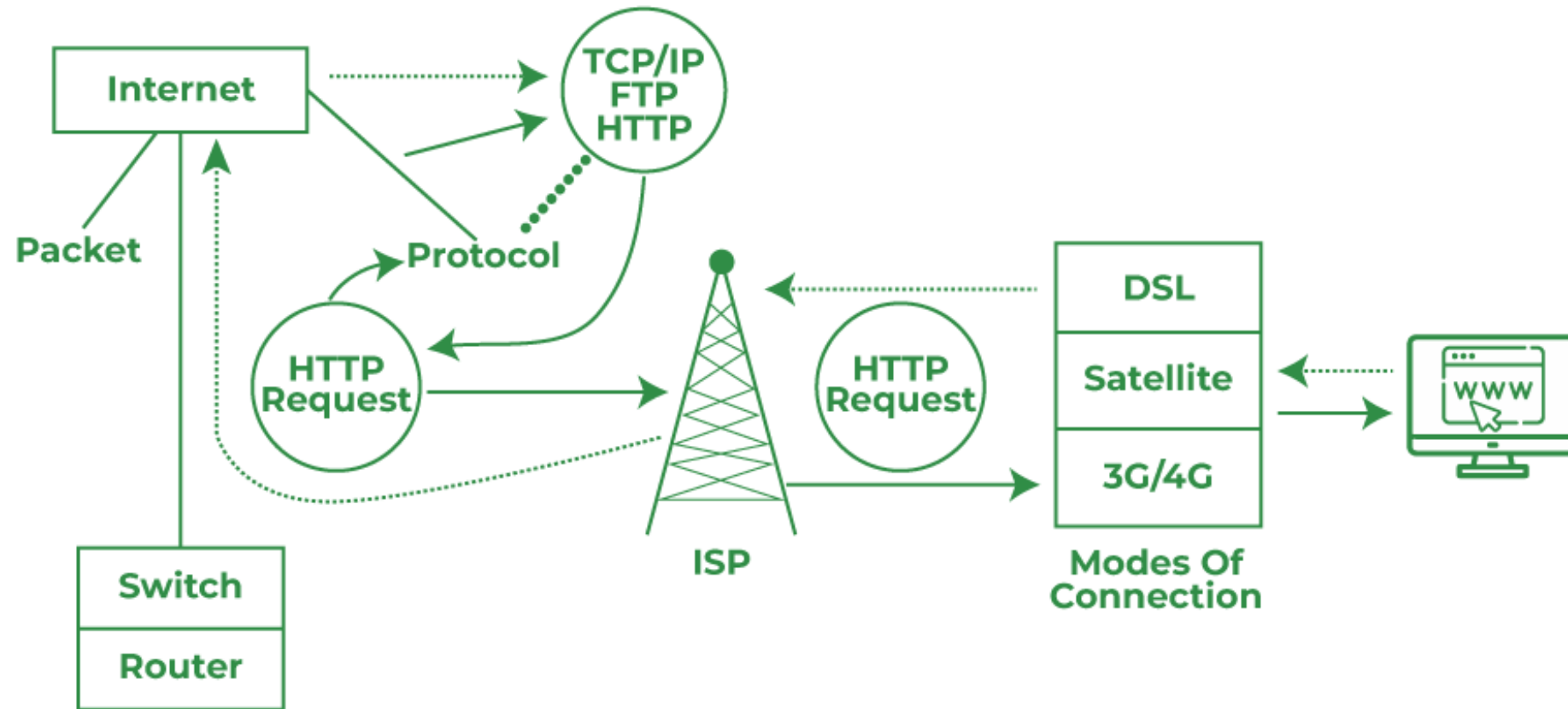


What is Web?

- The **World Wide Web**—usually called the **Web** for short—is a collection of different **websites** you can access through the Internet.
- A **website** is made up of related text, images, and other resources.
- Websites can resemble other forms of media—like newspaper articles or television programs—or they can be interactive in a way that's unique to computers.
- Once you are connected to the Internet, you can access and view websites using a type of application called a **web browser**.
- **Remember**, the web browser itself is not the Internet; it only displays websites that are stored on the Internet.



How does the Internet work?

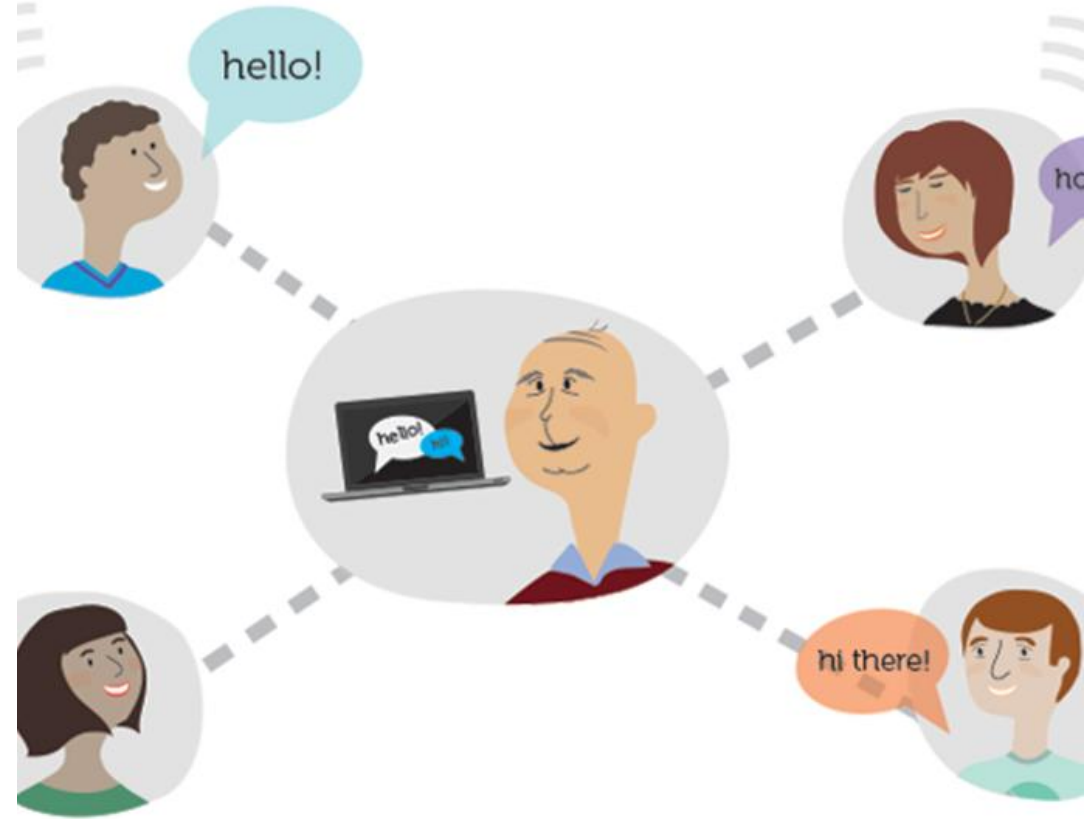


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- Firstly, you'll be required to connect your system or PC with any router or modem to establish a connection. This connection is the base of the internet connection.
- When you open the browser and start typing something like "www.google.com", your system will push a query command to your ISP (Internet Service Provider) that is connected with other servers that store and process data.
- Now, the web browser will start indexing the URL that you've entered and will fetch the details in numeric format (in their language to identify the address *unique*) that you're trying to reach.
- Next, now your browser will start sending the HTTP request where you're trying to reach and send a copy of the website on the user's system. Once all the data (of small packets) is received at the user's end (PC/Laptop), the browser will start arranging all those small packets and later will form a collective file (here, the browser will gather all the small packets and rearrange them just like a puzzle) and then you'll be able to see the contents of that website.

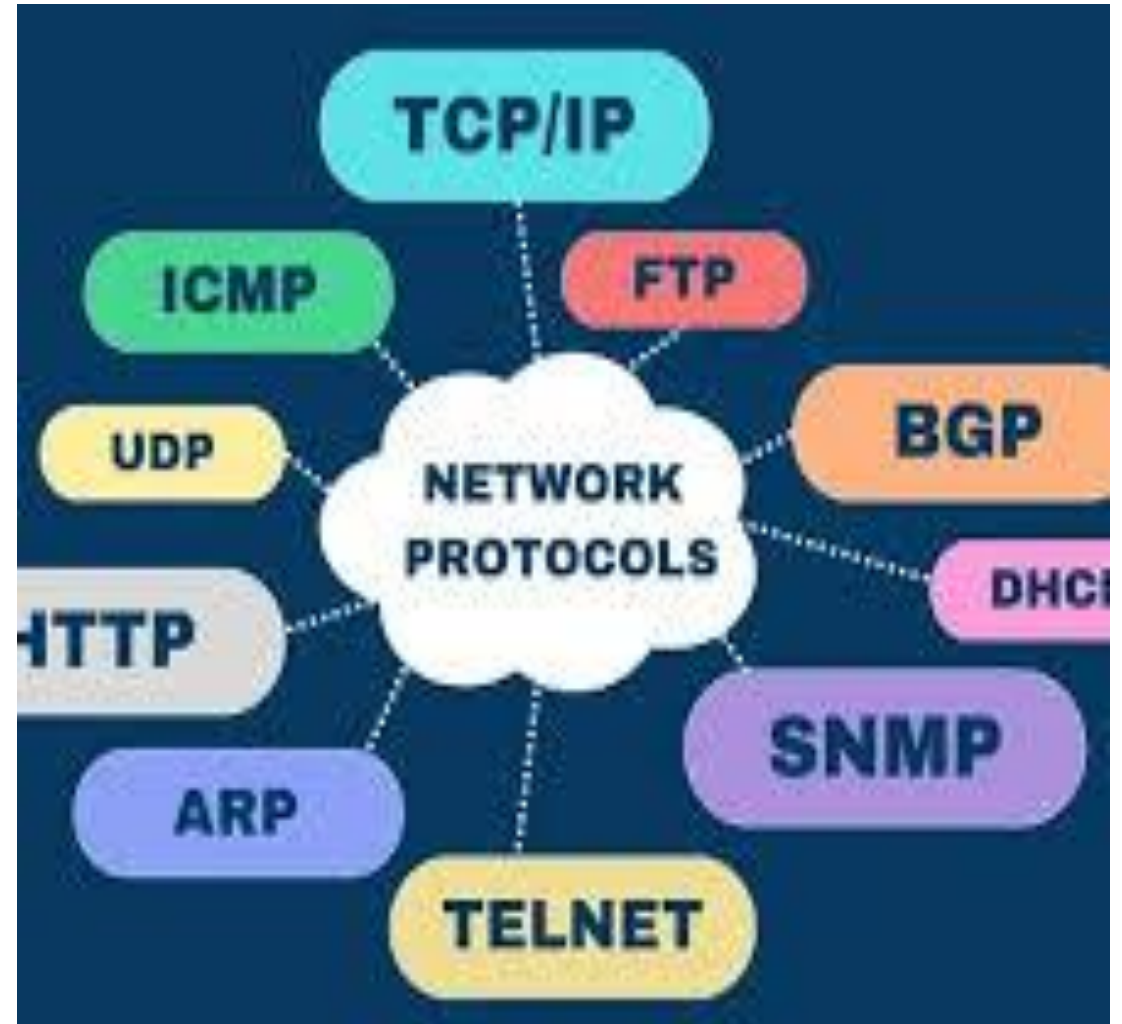
Other things you can do on the Internet

- ✓ Email
- ✓ Social Media
- ✓ Shopping
- ✓ Manage Your Bank accounts
- ✓ Watch TV

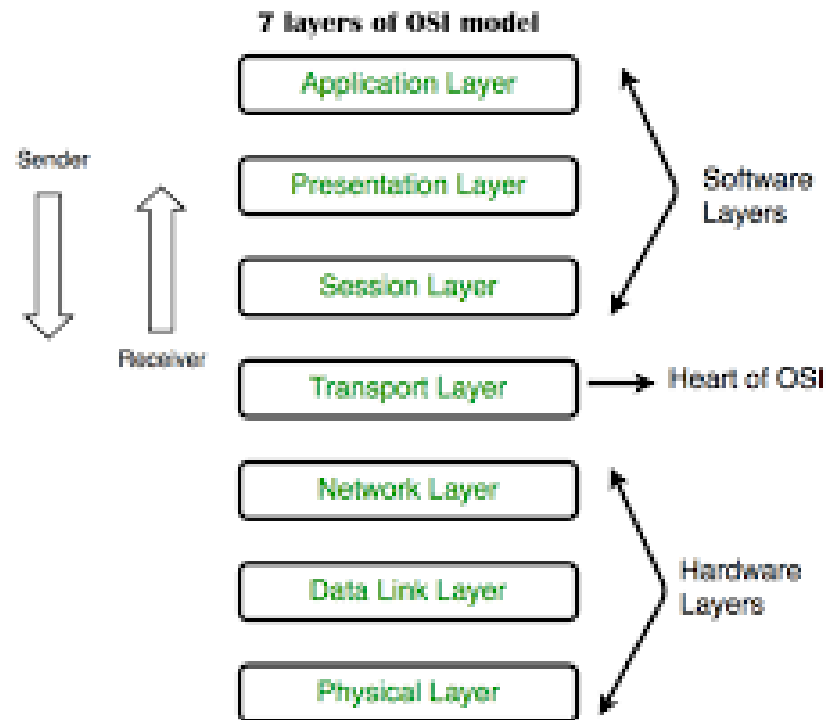


Network Protocols

- A protocol is a set of rules for formatting and processing data. Network protocols are like a common language for computers.
- The computers within a network may use vastly different software and hardware; however, the use of protocols enables them to communicate with each other.
- Standardized protocols are like a common language that computers can use, similar to how two people from different parts of the world may not understand each other's native languages, but they can communicate using a shared third language.
- On the Internet, there are different protocols for different types of processes. Protocols are often discussed in terms of which OSI model layer they belong to



Layers of OSI Model



Protocols run on the network layer

- IP is a network layer protocol responsible for routing. But it is not the only network layer protocol.
- **IPsec:** Internet Protocol Security (IPsec) sets up encrypted, authenticated IP connections over a [virtual private network \(VPN\)](#). Technically IPsec is not a protocol, but rather a collection of protocols that includes the Encapsulating Security Protocol (ESP), Authentication Header (AH), and Security Associations (SA).
- **ICMP:** The Internet Control Message Protocol (ICMP) reports errors and provides status updates. For example, if a router is unable to deliver a packet, it will send an ICMP message back to the packet's source.
- **IGMP:** The Internet Group Management Protocol (IGMP) sets up one-to-many network connections. IGMP helps set up multicasting, meaning multiple computers can receive data packets directed at one [IP address](#)

Other protocols are used on the Internet

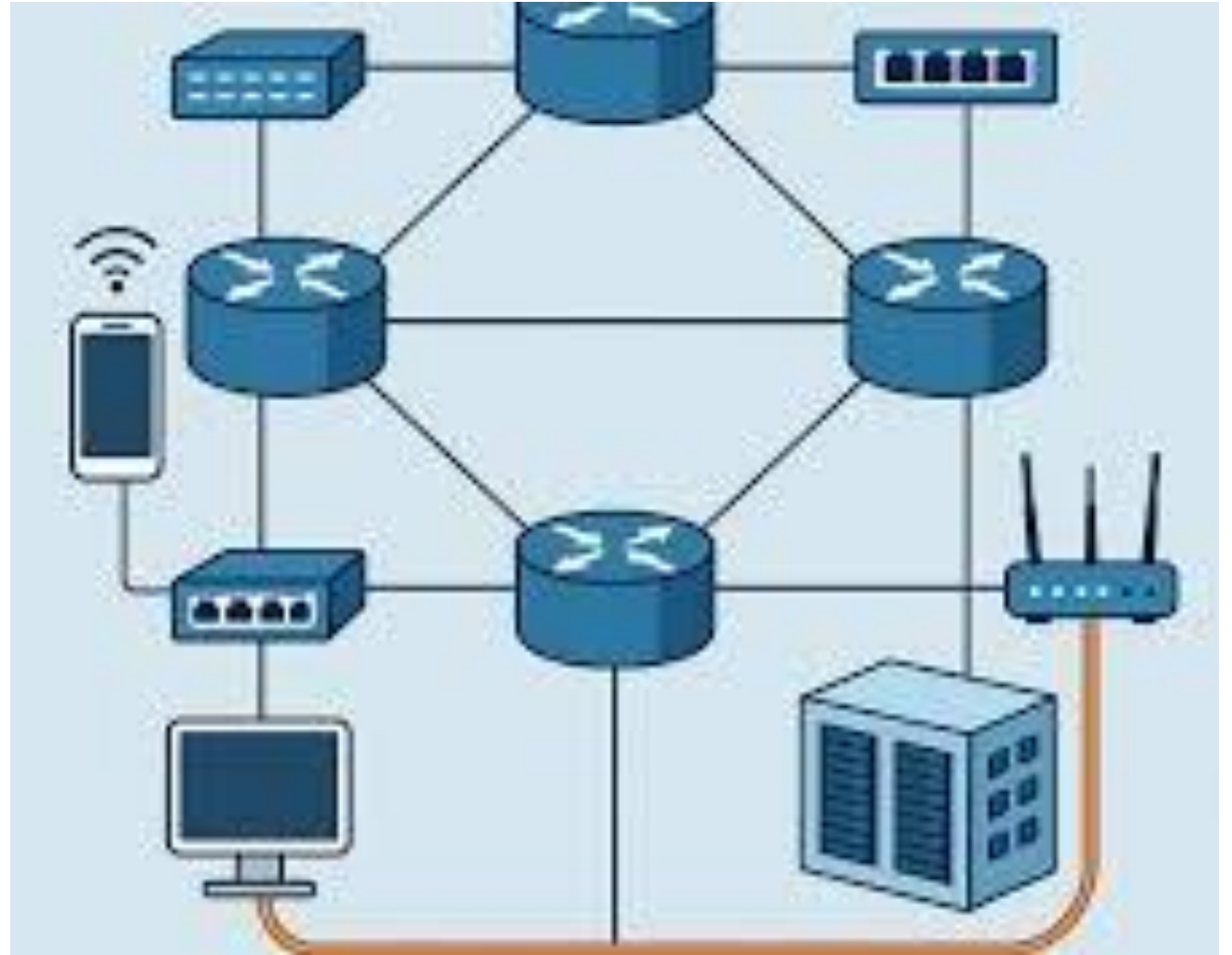
- **TCP:** TCP is a transport layer protocol that ensures reliable data delivery. TCP is meant to be used with IP, and the two protocols are often referenced together as TCP/IP.
- **HTTP:** The [Hypertext Transfer Protocol\(HTTP\)](#) is the foundation of the World Wide Web, the Internet that most users interact with. It is used for transferring data between devices. HTTP belongs to the application layer(layer 7), because it puts data into a format that applications (e.g. a browser) can use directly, without further interpretation. The lower layers of the OSI model are handled by a computer's operating system, not applications.
- **HTTPS:** The problem with HTTP is that it is not [encrypted](#) — any attacker who intercepts an HTTP message can read it.**HTTPS** (HTTP Secure) corrects this by encrypting HTTP messages.
- **TLS/SSL:** [Transport Layer Security\(TLS\)](#) is the protocol HTTPS uses for encryption. TLS used to be called [Secure Socket Layer\(SSL\)](#).
- **UDP:** The [User Datagram Protocol\(UDP\)](#) is a faster but less reliable alternative to TCP at the transport layer. It is often used in services like video streaming and gaming, where fast data delivery is paramount.

What protocols do routers use?

- **BGP:** The [Border Gateway Protocol \(BGP\)](#) is an application layer protocol networks use to broadcast which IP addresses they control. This information allows routers to decide which networks data packets should pass through on the way to their destinations.
- **EIGRP:** The [Enhanced Interior Gateway Routing Protocol \(EIGRP\)](#) identifies distances between routers. EIGRP automatically updates each router's record of the best routes (called a routing table) and broadcasts those updates to other routers within the network.
- **OSPF:** The [Open Shortest Path First \(OSPF\)](#) protocol calculates the most efficient network routes based on a variety of factors, including distance and bandwidth.
- **RIP:** The [Routing Information Protocol \(RIP\)](#) is an older routing protocol that identifies distances between routers. RIP is an application layer protocol.

Network Core

The **Network Core** is the central part of a network that interconnects various sub-networks, facilitating high-speed data transmission across long distances.



Key Characteristics

- **1. High-Speed Data Transmission:** The primary function of the network core is to transmit a **large** volume of data at very high speeds with minimal delay (low latency).

Why is it important?

- Millions of users access the Internet simultaneously.
- The network core must handle huge amounts of traffic without slowing down.

Example

- When you stream a 4K YouTube video, the video data travels through the network core at high speed, ensuring smooth playback without buffering.

Technologies Used

- Fiber optic cables
- High-speed routers
- Optical communication links

Continued...

2. Routing and Switching: The network core uses advanced routers and switches to determine the best path for data packets.

Routing

- Determines the best path from the source to the destination.
- Performed by routers.

Switching

- Transfers packets from one device or interface to another within the network.
- Performed by switches.

Example

- Suppose you send an email from Pune to Delhi.
- The routers in the network core decide the shortest and fastest route so that the email reaches its destination quickly.

Continued...

3. Protocol Support

- The network core uses various routing protocols to exchange routing information and efficiently forward packets.

(a) OSPF (Open Shortest Path First)

- An Interior Gateway Protocol (IGP) used within a single organization or Autonomous System (AS).
- Calculates the shortest path using Dijkstra's Algorithm.
- Provides fast convergence.

Example: Used inside a university campus or corporate network.

b) BGP (Border Gateway Protocol)

- An Exterior Gateway Protocol (EGP).
- Connects different Autonomous Systems (AS) on the Internet.
- Decides the best route between Internet Service Providers (ISPs).
- **Example:** Used by ISPs like Jio, Airtel, and BSNL to exchange routing information.

Continued...

4. Redundancy and Reliability

Definition

- The network core contains multiple communication paths between routers so that communication continues even if one path fails.

Redundancy

- Having more than one path between network devices.

Reliability

- Ensures continuous communication with minimal downtime.

Example

- If a fiber optic cable is damaged due to construction work, routers automatically redirect packets through another available path. Users may not even notice the failure.

Benefits

- High availability
- Fault tolerance
- Continuous service
- Disaster recovery

Examples of Network Core

1. Internet Backbone Infrastructure

- The main high-speed network connecting countries, cities, ISPs, and data centers.
- Built using high-capacity fiber optic cables.

Example: Submarine fiber optic cables connecting India with other countries.

2. Data Center Interconnections

- Large cloud providers connect multiple data centers using high-speed core networks.

Examples

- Google Cloud
- Microsoft Azure
- Amazon Web Services (AWS)
- This enables users to access cloud services quickly and reliably.

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3. Service Provider Core Networks

- Internet Service Providers (ISPs) maintain core networks to connect millions of customers.

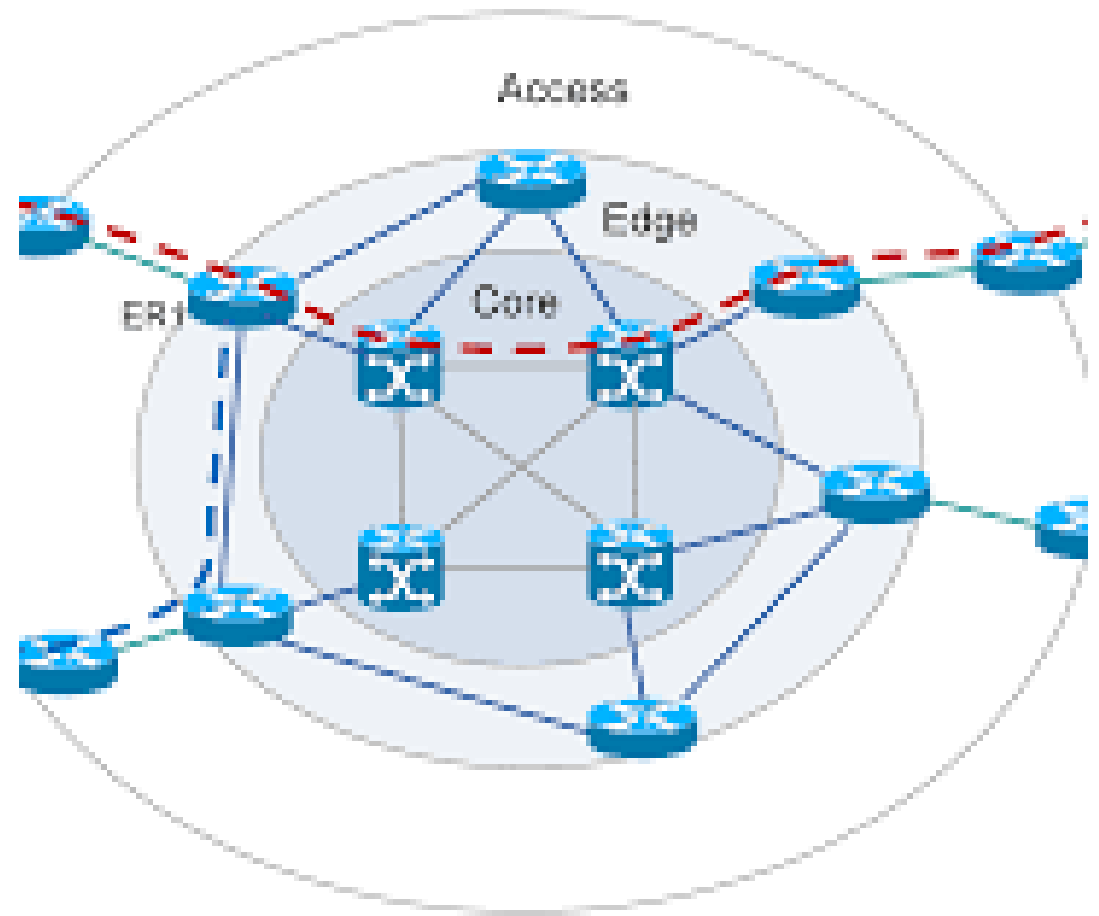
Examples

- Jio
- Airtel
- BSNL
- Vodafone Idea

These core networks route Internet traffic across cities and countries.

Network Edge

- The **Network Edge** refers to the boundary between the internal network and external networks, such as the internet. It's where end-user devices connect to the broader network infrastructure.



Continued...

1. Access Layer (Top)

- This is where **end-user devices** connect.
- Examples:
 - Laptops
 - Smartphones
 - Desktop computers
 - Printers
 - IoT devices
- Devices connect through **Wi-Fi Access Points, Ethernet switches, or routers.**

2. Edge Layer (Middle Ring)

- This is the **Network Edge**.
- It contains **edge routers and switches**.
- Responsibilities include:
 - Connecting local users to the Internet.
 - Forwarding packets toward the core network.
 - Performing security functions such as **firewalls, NAT, VPNs, and traffic filtering.**
 - Managing traffic entering and leaving the organization.

Continued...

3.Core Layer (Center)

- The **Core Network** is the backbone of the Internet.
- It consists of high-speed routers that move data between different networks.
- Its primary goal is fast and reliable packet forwarding.

How Data Travels

- Suppose you open **www.google.com** on your laptop.
- The data follows this path:
- **Laptop → Wi-Fi Router (Access Layer) → Edge Router (Network Edge) → Core Internet Network → Google's Server**
- When Google sends a response:
- **Google's Server → Core Network → Edge Router → Wi-Fi Router → Laptop**
- The **Network Edge** is the gateway through which all user traffic enters and exits the Internet.

Real-Life Example

- Imagine your college network:
- Your laptop connects to the college Wi-Fi.
- The Wi-Fi access point connects to an edge router.
- The edge router connects to the Internet Service Provider (ISP).
- The ISP forwards your request across the Internet to websites like Google or YouTube.

Here:

- Laptop → End device
- College Wi-Fi Router → Access Layer
- College Edge Router → Network Edge
- ISP Network → Core Network

Key Characteristics

1. Access Points: Access points are the devices that allow users and local networks to connect to the Internet or another network.

Devices Included

- Routers
- Gateways
- Firewalls
- Wireless Access Points (APs)
- Edge Switches

Functions

- Connect end-user devices to the network.
- Forward incoming and outgoing data packets.
- Act as the entry and exit point of the network.

Continued...

2. Security and Policy Enforcement:The network edge protects the internal network from unauthorized users and cyberattacks.

Security Mechanisms

- Firewall
- Intrusion Detection System (IDS)
- Intrusion Prevention System (IPS)
- VPN Gateway
- Access Control Lists (ACLs)

Functions

- Blocks unauthorized access.
- Filters malicious traffic.
- Monitors suspicious network activity.
- Enforces organizational security policies.

Example

- Suppose a hacker tries to access a company's internal server.
- The **firewall at the network edge** checks whether the request is authorized.
- If authorized → Traffic is allowed.
- If unauthorized → Traffic is blocked.

Continued...

3. Traffic Management: Traffic management ensures that network resources are used efficiently.

It performs tasks like:

(a) Quality of Service (QoS)

- QoS gives higher priority to important traffic.

Example

During an online lecture:

- Video conferencing traffic gets high priority.
- File downloads get lower priority.
- This ensures smooth audio and video without interruptions.

(b) Load Balancing

- Load balancing distributes network traffic across multiple servers or links.

Benefits

- Prevents congestion.
- Improves performance.
- Increases availability.
- Reduces response time.

Example

A website like Amazon receives millions of requests every minute. Instead of one server handling everything, requests are distributed across many servers.

Continued...

4. Edge Computing: Edge computing means processing data close to where it is generated instead of sending everything to a distant cloud.

Benefits

- Lower latency
- Faster response time
- Reduced bandwidth usage
- Better real-time performance

Example

- A smart traffic camera detects vehicleS.

Continued...

- **Enterprise perimeter routers** act as the gateway between an organization's internal network and the external internet, providing a crucial first line of defense against cyber threats.
- They are typically the last router under the organization's control before traffic enters or exits the public network.
- **A content delivery network (CDN)** is a group of geographically distributed servers that speed up the delivery of web content by bringing it closer to where users are.
- **Edge computing in IoT deployments** involves processing data closer to the source, using devices like sensors, gateways, and routers, rather than relying solely on a central cloud.
- This approach reduces latency, enhances efficiency, and improves overall performance by minimizing the need to send all data to a remote cloud for processing.
- Edge computing also offers benefits like improved security, reduced bandwidth consumption, and the ability to handle real-time data processing in scenarios with limited or no cloud connectivity.

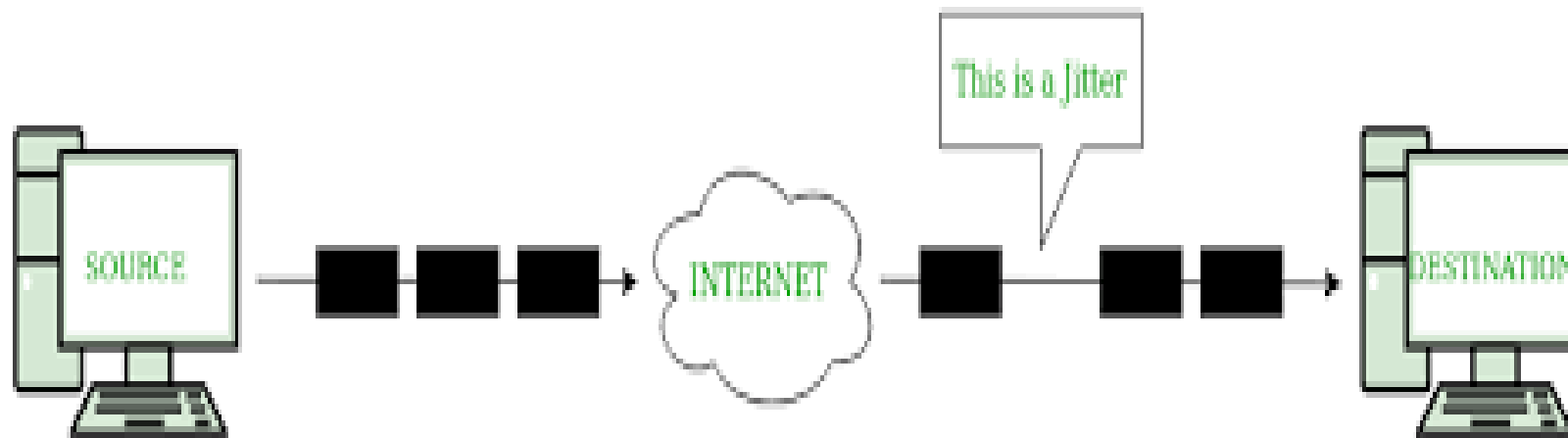
Difference Between Network Edge and Network Core

Feature	Network Edge	Network Core
Location	Boundary of the network	Center (backbone) of the network
Main Devices	PCs, laptops, smartphones, routers, switches	High-speed routers
Purpose	Connect users to the network	Transfer data between networks
Speed	Moderate	Very high
Security	Firewalls, NAT, VPN	Mainly routing and forwarding

Performance Measures:

The performance of a network pertains to the measure of service quality of a network as perceived by the user.

There are different ways to measure the performance of a network, depending upon the nature and design of the network. Finding the performance of a network depends on both quality of the network and the quantity of the network.



Delay(Latency)

Definition: Delay refers to the time it takes for a data packet to travel from the source to the destination across a network. It is typically measured in milliseconds (ms) and can be influenced by various factors such as distance, network congestion, and the quality of network equipment.

Components of Delay:

- **Processing Delay:** Time taken by routers or switches to process packet headers, check for errors, and determine the next hop.
- **Queuing Delay:** Time a packet spends waiting in routing queues due to network congestion.
- **Transmission Delay:** Time required to push all the packet's bits onto the link, influenced by the packet's size and the link's bandwidth.
- **Propagation Delay:** Time for a signal to propagate through the transmission medium, dependent on the distance and the propagation speed of the medium.

3. Transmission Delay

Definition

Transmission delay is the time required to place (push) all bits of the packet onto the transmission link.

It depends on:

- Packet size
- Link bandwidth

Formula

$$\text{Transmission Delay} = \frac{\text{Packet Size (bits)}}{\text{Bandwidth (bits/sec)}}$$

Example

Packet Size = 1000 bits

Bandwidth = 1 Mbps

$$\frac{1000}{1,000,000} = 0.001 \text{ sec} = 1 \text{ ms}$$

Transmission delay = 1 ms

4. Propagation Delay

Definition

Propagation delay is the time taken by the signal to travel through the physical medium (copper cable, fiber optic, or wireless).

It depends on:

- Distance between sender and receiver
- Speed of signal in the medium

Formula

$$\text{Propagation Delay} = \frac{\text{Distance}}{\text{Propagation Speed}}$$

Example

Distance = 300 km

Propagation Speed = 2×10^8 m/s

$$\frac{300000}{2 \times 10^8} = 0.0015 \text{ sec} = 1.5 \text{ ms}$$

Propagation delay = 1.5 ms

Total Delay Example

Suppose:

- Processing Delay = **1 ms**
- Queuing Delay = **3 ms**
- Transmission Delay = **2 ms**
- Propagation Delay = **4 ms**

Then,

- Total Delay = $1 + 3 + 2 + 4 = 10$ ms

So, the packet takes **10 milliseconds** to reach the destination.

Real-Life Analogy

Think of sending a parcel through a courier service:

- **Processing Delay:** Courier staff check and label the parcel.
- **Queuing Delay:** Parcel waits with other parcels before dispatch.
- **Transmission Delay:** Parcel is loaded onto the truck.
- **Propagation Delay:** Truck travels from one city to another.

The total delivery time is the sum of all these stages, just as network latency is the sum of all four delays.

Packet Loss

Definition: Packet loss occurs when data packets fail to reach their destination, often due to network congestion, faulty hardware, or software bugs.

Causes:

- **Network Congestion:** When the network is overwhelmed with traffic, routers and switches may drop packets to manage the load.
- **Hardware Failures:** Defective network devices can lead to packet loss.
- **Buffer Overflow:** If network buffers are full, incoming packets may be discarded.

Packet loss is usually expressed as a **percentage (%)**.

Example

Suppose a sender transmits **100 packets** to a receiver.

- 98 packets arrive successfully.
- 2 packets are lost.

Then,

$$\text{Packet Loss} = \frac{2}{100} \times 100 = 2\%$$

Packet Loss = 2%

Causes of Packet Loss

1. Network Congestion

- **Definition**
- When too many packets travel through the network at the same time, routers and switches become overloaded.
- Since routers have limited memory (buffer), they start **dropping packets**.

Example

- Imagine a highway during rush hour.
- Too many vehicles enter the road.
- Traffic becomes congested.
- Some vehicles cannot move efficiently.
- Similarly, when network traffic is very high, some packets are discarded.

Real-life Example

- During online exams or live sports streaming, thousands of users access the network simultaneously, causing congestion and packet loss.

Continued...

2. Hardware Failures

Definition

- Packet loss can occur if network devices are damaged or malfunctioning.
- Examples include:
 - Faulty routers
 - Damaged switches
 - Broken Ethernet cables
 - Defective Network Interface Cards (NICs)

Example

- If a router's hardware fails, it may be unable to forward packets, causing them to be lost.

Continued...

3. Buffer Overflow

- **Definition**
- Routers temporarily store incoming packets in a **buffer (memory)** before forwarding them.
- If packets arrive faster than they can be transmitted, the buffer becomes full.
- When the buffer is full, new incoming packets are discarded.
- **Example**
- Suppose a router's buffer can hold **100 packets**.
- 120 packets arrive at the same time.
- The first 100 are stored.
- The remaining 20 are dropped.
- This is called **Buffer Overflow**.

Packet Loss vs Delay

Packet Loss

Packets are completely lost.

Measured in percentage (%).

Causes retransmissions and missing data.

Often caused by congestion, hardware failure, or buffer overflow.

Delay (Latency)

Packets reach the destination but arrive late.

Measured in milliseconds (ms).

Causes slow communication.

Caused by processing, queuing, transmission, and propagation delays.

Throughput

- **Throughput** is the actual amount of data that is successfully transferred from the source to the destination per unit time over a network.
- It represents the real data transfer rate achieved by the network.
- It is usually measured in:
 - bits per second (**bps**)
 - Kilobits per second (**Kbps**)
 - Megabits per second (**Mbps**)
 - Gigabits per second (**Gbps**)

Example

Suppose a network transfers 500 MB of data in 10 seconds.

$$\begin{aligned}\text{Throughput} &= \frac{\text{Data Transferred}}{\text{Time}} \\ &= \frac{500 \text{ MB}}{10} = 50 \text{ MB/s}\end{aligned}$$

Therefore, the throughput is 50 MB/s.

Factors Influencing Throughput:

- **Bandwidth**
- **Latency**
- **Packet Loss**
- **Network Congestion**

CONTINUED...

1. Bandwidth

Definition

- **Bandwidth** is the maximum capacity of a communication link, i.e., the maximum amount of data that can theoretically be transmitted per second.
- Higher bandwidth generally allows higher throughput.

Example

- Network A: 10 Mbps bandwidth
- Network B: 100 Mbps bandwidth
- The 100 Mbps network can transfer much more data in the same amount of time.

Continued...

2. Latency (Delay)

Definition

- Latency is the time taken for a packet to travel from the sender to the receiver.
- Higher latency reduces throughput because packets take longer to reach the destination.

Example

Suppose:

- Satellite Internet → High latency (600 ms)
- Fiber Internet → Low latency (20 ms)
- Even if both have the same bandwidth, the fiber network usually provides better throughput because data reaches the destination faster.

Continued...

3. Packet Loss

Definition

- When packets are lost during transmission, they must be retransmitted.
- Retransmissions consume extra bandwidth and reduce the actual throughput.

Example

- Suppose 100 packets are sent.
- 95 packets arrive successfully.
- 5 packets are lost.
- The sender retransmits the lost packets.
- This additional transmission reduces the effective throughput.

Continued...

4. Network Congestion

Definition

- When many users share the same network simultaneously, the network becomes congested.
- Congestion causes:
- Longer waiting time
- Packet loss
- Higher delay
- As a result, throughput decreases.

Example

- Imagine hundreds of students using the college Wi-Fi during lunch break.
- Since many devices share the same bandwidth, everyone experiences slower download speeds.